

Topic: Interrupt Structure and its Servicing -- Self-Learning

Q. Discuss the Interrupt structure of the 8086 microprocessor. Differentiate between Hardware and Software interrupts, and explain the sequence of steps the CPU follows to service an interrupt.

[DEF] Definition to Remember

An Interrupt pauses the CPU's normal execution to handle an urgent task (an Interrupt Service Routine or ISR). The 8086 supports 256 interrupts. It uses an Interrupt Vector Table (IVT) in lower memory to locate the ISR addresses, ensuring the CPU state is saved on the stack before jumping.

Interrupt Structure & Servicing in 8086 (Self-Learning)

Type	Source	Characteristics	Examples
Hardware	External physical devices	Triggered by electrical signals to CPU	**INTR** (Maskable), **NMI** (Non-Maskable)
Software	Internal program execution	Triggered by the `INT n` instruction or exceptions	`INT 21H` (DOS), Divide-by-zero (Type 0)

2. The Interrupt Vector Table (IVT)

* Located at absolute memory 00000H to 003FFH.

3. Interrupt Servicing Sequence

When a valid interrupt is acknowledged, the 8086 performs the following strict sequence to ensure it doesn't lose its place in the main program:

[MUST-WRITE] Must-Write Points (for 10 marks)

1. An Interrupt forces the CPU to pause execution and run a specific routine (ISR).
2. Hardware Interrupts: External signals. INTR is maskable; NMI is non-maskable (for critical errors).
3. Software Interrupts: Triggered by INT n instructions or exceptions (divide-by-zero).
4. Interrupt Vector Table (IVT): Occupies the lowest 1KB of memory (00000H-003FFH).
5. IVT holds 256 entries. Each entry is 4 bytes (CS and IP for the ISR).
6. Servicing Sequence: Push Flags -> Clear IF/TF -> Push CS and IP -> Load new CS/IP from IVT -> Execute ISR.
7. The ISR must end with IRET to pop the IP, CS, and Flags from the stack and resume the program.

[RECALL] Quick Recall

Interrupts -> Hardware (INTR/NMI) vs Software (INT n) -> IVT (Lowest memory, 256 types, 4 bytes each for CS:IP) -> Sequence: Push Flags -> Clear IF/TF -> Push CS:IP -> Fetch ISR from IVT -> Run ISR -> IRET (Pop all back)